

Tower Generation from Möbius Holonomy: Canonical Resolution of Representational Obstruction

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Abstract

We show that the Möbius connection $\omega = \frac{1}{2}d\theta$ on S^1 generates an infinite canonical tower of algebraic and geometric structures through iterated resolution of representational obstruction. At each level, the subobject classifier of the associated topos contains *swap* values—propositions that are locally definite but globally unresolvable. These constitute cohomological obstructions whose minimal isometric resolution is unique and forced, producing the Cayley–Dickson doubling of the current algebra.

The first four levels recover the normed division algebras $\mathbb{R}, \mathbb{C}, \mathbb{H}, \mathbb{O}$. Beyond $\mathbb{O}$, zero divisors appear and the resolution outputs constraint geometry rather than division algebras. We prove three structural theorems for the post-division-algebra regime: (1) a *two-sidedness theorem* establishing that zero divisor relations among pure imaginary elements are symmetric, derived from the linearized norm identity and flexibility of Cayley–Dickson algebras; (2) a *degree-doubling theorem* showing that every zero-divisor node exactly doubles its connectivity at each level of the tower; and (3) a *non-termination theorem* establishing that the tower generates strictly new structure at every level. Together, these establish that the tower is the unique canonical object generated by iterating the Möbius connection’s holonomy through its own representational obstructions, with no choice at any step.

Explicit computation through six post-NDA levels (dimension 16 to 512, over eight million zero-divisor pairs) reveals unexpectedly rigid structure: the strata count follows $2^n - 1$ exactly, singular values of multiplication maps are restricted to $\{0, 1, \sqrt{2}\}$ at every level, kernel dimensions satisfy $\dim(\ker L_a) = 2 \cdot \deg(a)$ for every element, and the growth ratio of zero-divisor pairs converges to $8 = \dim(\mathbb{O})$. The constraint geometry exhibits complete structural inheritance, mirror symmetry with shelf structure in the degree distributions, and a chirality decomposition into self-adjoint and chiral-paired components with exact count formulas. All stated laws hold with zero exceptions across the entire dataset.

The spectral rigidity is identified as the three-valued subobject classifier of Paper III propagating through the tower: the singular values $\{0, 1, \sqrt{2}\}$ are the norms of the membership vectors on the two-sheeted Möbius cover, the squared norms yield a Born rule (observable = amplitude squared), and the transformation between the holonomy eigenbasis and the sheet basis is the Hadamard gate. The *swap* truth value is a coherent superposition of the ± 1 holonomy eigenvalues, preserved without decoherence at every level.

This paper is the fourth in a series. Paper I [1] establishes the Möbius connection as the canonical generator of arithmetic structure. Paper II [2] derives consequences for the Riemann Hypothesis via orientation and fixed-point loci. Paper III [3]

derives the three-valued subobject classifier of $\mathbb{Z}_2\text{-Set}$ and resolves set-theoretic paradoxes.

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1 Introduction

The Möbius connection $\omega = \frac{1}{2}d\theta$ on S^1 is the unique flat connection whose holonomy group is $\mathbb{Z}_2$ and whose associated double cover is connected [1]. Papers I–III established three consequences of this object:

- (i) The connection canonically generates $\mathbb{C}$ as a degree-2 field extension of $\mathbb{R}$, with complex conjugation as the deck transformation of the double cover (Paper I).
- (ii) The fixed-point loci of the identification structures on the canonical $\mathbb{C}$ determine the critical line $\operatorname{Re}(s) = \frac{1}{2}$ as the unique vanishing locus for equivariant sections (Paper II).
- (iii) The holonomy group $\mathbb{Z}_2$ generates the topos $\mathbb{Z}_2\text{-Set}$, whose subobject classifier $\Omega = \{\mathbf{true}, \mathbf{false}, \mathbf{swap}\}$ provides a three-valued logic in which classical reasoning holds on the fixed-point locus and fails at the **swap** values (Paper III).

The present paper asks: *what happens to the swap values?*

The **swap** truth value classifies propositions that are locally definite but globally orientation-dependent. We show that this irresolvability is not a terminal condition but a *generative* one: the swap values constitute an obstruction whose minimal resolution forces the existence of the next algebraic level, and this forcing is canonical—unique at each step, with the output determined without remainder.

Following the resolution through four levels recovers the normed division algebras $\mathbb{R}, \mathbb{C}, \mathbb{H}, \mathbb{O}$ as forced consequences. Continuing past the division algebra boundary reveals organized constraint geometry—zero-divisor structures with specific combinatorial properties, controlled growth, and complete inheritance of lower-level structure. The central results of this paper are:

- (1) A *canonicity theorem* establishing that the Cayley–Dickson doubling is the unique resolution of the swap obstruction at each level (Theorem 4.1).
- (2) A *two-sidedness theorem* for zero divisors, derived algebraically from the linearized norm identity and flexibility (Theorem 5.4).
- (3) A *degree-doubling theorem* showing that every inherited zero-divisor node exactly doubles its connectivity (Theorem 5.7).
- (4) A *non-termination theorem* establishing that the tower generates strictly new structure at every level (Theorem 6.1).

2 The Swap Value as Cohomological Obstruction

2.1 The three-valued classifier

We recall from Paper III the subobject classifier of $\mathbb{Z}_2\text{-Set}$.

Proposition 2.1 (Paper III). *The subobject classifier of $\mathbb{Z}_2\text{-Set}$ is $\Omega = (\{\mathbf{true}, \mathbf{false}, \mathbf{swap}\}, \sigma_\Omega)$. For any $\mathbb{Z}_2$ -set (X, σ) and equivariant subset $A \hookrightarrow X$, the characteristic morphism $\chi_A: X \rightarrow \Omega$ assigns:*

$$\begin{array}{ll} \chi_A(x) = \mathbf{true} & \text{if } x \in A \text{ and } \sigma(x) \in A, \\ \chi_A(x) = \mathbf{false} & \text{if } x \notin A \text{ and } \sigma(x) \notin A, \\ \chi_A(x) = \mathbf{swap} & \text{if exactly one of } x, \sigma(x) \text{ belongs to } A. \end{array}$$

2.2 Cohomological interpretation

Proposition 2.2 (Swap as non-trivial cocycle). *Define $c_A: X \rightarrow \mathbb{Z}_2$ by $c_A(x) = 1$ if $x \in A$, $c_A(x) = 0$ otherwise. Then $\chi_A(x) = \text{swap}$ if and only if the $\mathbb{Z}_2$ -coboundary δc_A is nonzero on the orbit containing x : specifically, $\delta c_A(\{x, \sigma(x)\}) = c_A(\sigma(x)) - c_A(x) \neq 0 \pmod{2}$.*

Proof. Direct verification: $\delta c_A \neq 0$ on the orbit $\{x, \sigma(x)\}$ if and only if exactly one of $x, \sigma(x)$ belongs to A , which is the definition of $\chi_A(x) = \text{swap}$. $\square$

The swap class cannot be removed by any equivariant modification of A . It is a topological obstruction reflecting the non-orientability of the underlying Möbius bundle.

Remark 2.3 (Scope of “cohomological”). Throughout this paper, “cohomological” is used in the elementary sense of $\mathbb{Z}_2$ -valued group cohomology on orbits: the swap value corresponds to a non-trivial $\mathbb{Z}_2$ cocycle on the orbit space. No higher derived-functor machinery, sheaf cohomology, or spectral sequence arguments are invoked. The full cohomological formulation—in which the obstruction theory of Section 2 is embedded in the internal cohomology of the tower topos—is deferred to subsequent work.

3 The Division Algebra Levels

3.1 Level one: $\mathbb{R} \rightarrow \mathbb{C}$

The Möbius connection generates a $\mathbb{Z}_2$ holonomy that cannot be represented within $\mathbb{R}$. The holonomy acts by sign reversal: $x \mapsto -x$. Within $\mathbb{R}$, this operation has a fixed point at 0 but no square root—there is no element $i \in \mathbb{R}$ with $i^2 = -1$ that would internalize the sign reversal as a rotation. The unique minimal field extension absorbing this holonomy as an internal operation (complex conjugation) is $\mathbb{C} = \mathbb{R}[i]/(i^2 + 1)$ (Papers I and II). The swap value at this level classifies propositions whose truth depends on sign convention—positive on one sheet, negative on the other. The extension to $\mathbb{C}$ resolves this by encoding both signs simultaneously as real and imaginary parts. This extension is degree 2 and unique up to isomorphism.

3.2 Level two: $\mathbb{C} \rightarrow \mathbb{H}$

At the $\mathbb{C}$ level, the $\mathbb{Z}_2$ swap can be applied along any axis in the imaginary plane, parametrized by S^1 . But once $\mathbb{C}$ is embedded in a higher structure, independent swaps can be assigned to different complex subplanes, now parametrized by S^2 . Concretely: if swap axis α resolves a proposition by assigning it phase $e^{i\theta}$, and swap axis β resolves a different proposition by assigning phase $e^{i\phi}$, the composite resolution depends on the *order* in which α and β are applied whenever $\alpha \neq \pm\beta$. This order-dependence cannot be represented in $\mathbb{C}$, where all multiplications commute. The minimal extension absorbing this non-commutative interference is $\mathbb{H}$, the unique 4-dimensional normed division algebra. The loss of commutativity is the algebraic cost of faithfully encoding transport-order dependence: $ij = k \neq -k = ji$.

3.3 Level three: $\mathbb{H} \rightarrow \mathbb{O}$

At the $\mathbb{H}$ level, pairs of transport operations compose unambiguously (quaternionic multiplication is associative). But compositions of three or more operations can depend on grouping: $(ab)c$ versus $a(bc)$. The *associator* $[a, b, c] = (ab)c - a(bc)$ measures this grouping-dependence. When three independent swap axes are composed, the associator is generically nonzero, and no quaternionic representation can absorb this without losing information. The minimal extension absorbing the associativity obstruction is $\mathbb{O}$, the unique 8-dimensional normed division algebra. The loss of associativity is the cost of encoding grouping-dependence. The octonions retain a weaker property, *alternativity* ($a(ab) = a^2b$ and $(ab)b = ab^2$), which suffices to preserve norm multiplicativity: $N(ab) = N(a)N(b)$.

The four NDAs $\mathbb{R}, \mathbb{C}, \mathbb{H}, \mathbb{O}$ are the only finite-dimensional normed division algebras over $\mathbb{R}$ (Hurwitz, 1898 [4]). The Cayley–Dickson construction, applied once more, produces the sedenions $\mathbb{S} = \text{CD}(\mathbb{O})$, which necessarily contain zero divisors—elements $a, b \neq 0$ with $ab = 0$. This is the entry point to the post-NDA regime.

4 The Canonicity Theorem

The tower generation principle claims that the resolution at each level is not merely natural but *unique*. We now make this precise by identifying every point where a choice could in principle enter, and showing that no choice is available.

Here and throughout, “canonical” and “unique” are understood in the sense established in Paper I: uniqueness relative to *oriented arithmetic structures* (OAS) generated by resolution of representational obstruction under the Möbius connection. The OAS category and its initial object theorem define the class within which the uniqueness claim operates.

4.1 Potential choice points

The Cayley–Dickson construction adjoins a new element ℓ to an algebra A with involution, producing $\text{CD}(A) = \{(a, b) : a, b \in A\}$ with multiplication

$$(a, b)(c, d) = (ac - \bar{d}b, da + b\bar{c}).$$

Three potential choice points arise:

P1: The sign of ℓ^2 . The new element $\ell = (0, 1)$ satisfies $\ell^2 = (-1, 0)$ by direct computation from the CD multiplication rule. The alternative $\ell^2 = +1$ would produce a *split* algebra with indefinite norm $N(a, b) = N(a) - N(b)$. Since the Möbius connection is an isometry (orientation reversal preserves distances), the involution on the doubled algebra must also be isometric, requiring positive-definite norm $N(a, b) = N(a) + N(b)$. This forces $\ell^2 = -1$.

P2: The multiplication rule. The CD multiplication formula is determined by four requirements:

- (R1) **Embedding:** $(a, 0)(c, 0) = (ac, 0)$ —the lower algebra embeds as a subalgebra.
- (R2) **New unit:** $\ell^2 = -1$ —forced by P1.

(R3) **Conjugation:** $(a, b)^* = (\bar{a}, -b)$ —the unique involution compatible with R1 and R2.

(R4) **Norm:** $N((a, b)) = N(a) + N(b)$ —forced by isometry.

Together, R1–R4 uniquely determine the multiplication formula. The CD construction is the unique doubling of a $*$ -algebra preserving these four properties.

P3: Whether doubling is the only resolution. The swap obstruction at each level is a $\mathbb{Z}_2$ identification that cannot be resolved within the existing algebra. Resolving a $\mathbb{Z}_2$ requires exactly one new degree of freedom. Three possibilities exist for the square of the new element: $\ell^2 = 0$ (nilpotent—collapses to a quotient, not an extension), $\ell^2 = +1$ (split—violates isometry), or $\ell^2 = -1$ (rotational—the CD construction). Only the rotational case is compatible with the Möbius connection’s isometric character.

Theorem 4.1 (Canonicity). *The Möbius connection $\omega = \frac{1}{2}d\theta$ uniquely determines an infinite tower of algebraic structures via the following chain of forced steps:*

- (C1) ω has $\mathbb{Z}_2$ holonomy (Paper I, unique).
- (C2) $\mathbb{Z}_2$ generates the topos $\mathbb{Z}_2\text{-Set}$ (standard, unique).
- (C3) $\mathbb{Z}_2\text{-Set}$ has the three-valued classifier $\{\mathbf{true}, \mathbf{false}, \mathbf{swap}\}$ (standard, forced).
- (C4) Swap values are cohomological obstructions—non-trivial $\mathbb{Z}_2$ -cocycles (Proposition 2.2).
- (C5) The minimal isometric resolution of a $\mathbb{Z}_2$ swap is adjunction of one imaginary unit ℓ with $\ell^2 = -1$ (P1 + P3 above).
- (C6) The multiplication rule is uniquely determined by R1–R4 (P2 above).
- (C7) $\text{CD}(A)$ inherits all of A ’s zero divisors and doubles their degree (Theorems 5.4 and 5.7).
- (C8) $\text{CD}(A)$ contains new zero-divisor structure not present in A , forcing the next step (Theorem 6.1).

At no step is there a choice. The properties lost at each level (commutativity at $\mathbb{H}$, associativity at $\mathbb{O}$, norm multiplicativity at $\mathbb{S}$) are consequences of the forced construction, not choices.

5 The Algebraic Engine: Two-Sidedness and Degree-Doubling

This section establishes the algebraic machinery that drives the tower past the division algebra boundary. The key results use only two standard properties of Cayley–Dickson algebras that hold at *every* level: the linearized norm identity and flexibility.

We fix notation. Let A denote a Cayley–Dickson algebra at level $n \geq 4$ (dimension 2^n), with conjugation $\bar{\cdot}$, norm $N(x) = x\bar{x}$, and real inner product $\langle x, y \rangle = \frac{1}{2}(x\bar{y} + y\bar{x})$. An element $a \in A$ is *pure imaginary* if $\bar{a} = -a$ (equivalently, $\langle a, 1 \rangle = 0$). The *degree* of a zero-divisor element a is the number of normalized basis-element sums b with $ab = 0$ and $b \neq 0$. The embedding $\iota: A \rightarrow \text{CD}(A)$ is $\iota(a) = (a, 0)$.

5.1 The linearized norm identity

Lemma 5.1 (Standard; see [5]). *In any Cayley–Dickson algebra, for all elements x, y :*

$$x\bar{y} + y\bar{x} = 2\langle x, y \rangle \cdot 1,$$

where $\langle x, y \rangle$ is the real inner product and $\bar{x}$ denotes conjugation. For pure imaginary elements (those with $\bar{x} = -x$), this specializes to:

$$xy + yx = -2\langle x, y \rangle \cdot 1. \quad (1)$$

Proof. Linearize the norm identity $x\bar{x} = N(x) \cdot 1$ by replacing x with $x + y$: $N(x + y) = (x + y)\overline{(x + y)} = x\bar{x} + x\bar{y} + y\bar{x} + y\bar{y} = N(x) + x\bar{y} + y\bar{x} + N(y)$. Since $N(x + y) = N(x) + 2\langle x, y \rangle + N(y)$, we obtain $x\bar{y} + y\bar{x} = 2\langle x, y \rangle \cdot 1$. For pure imaginary x, y : $\bar{x} = -x$, $\bar{y} = -y$, giving $-xy - yx = 2\langle x, y \rangle \cdot 1$. $\square$

5.2 Flexibility

Lemma 5.2 (Standard; see [5]). *All Cayley–Dickson algebras satisfy the flexibility identity: $x(yx) = (xy)x$ for all x, y .*

5.3 The two-sidedness theorem

The algebraic theorems in this section apply to pure imaginary elements—those satisfying $\bar{a} = -a$, equivalently those with zero real part. The following verified fact establishes that this hypothesis is not restrictive:

Proposition 5.3 (Pure imaginary hypothesis). *Among normalized sums of pairs of basis elements in the Cayley–Dickson algebras at dimensions 16 through 512, every zero-divisor node is pure imaginary. No element with nonzero real component participates in any zero-divisor pair.*

This is verified computationally (84/84 nodes at dimension 16, extending through all six levels with zero exceptions). An algebraic proof follows from the observation that the real unit $1 = e_0$ is invertible in every Cayley–Dickson algebra and therefore cannot be a zero divisor or contribute to one: if $a = \alpha \cdot 1 + a'$ with a' pure imaginary and $\alpha \neq 0$, then a is invertible by the norm identity.

Theorem 5.4 (Two-sidedness of zero divisors). *Let a, b be pure imaginary elements of a Cayley–Dickson algebra with $ab = 0$ and $b \neq 0$. Then $ba = 0$ and $\langle a, b \rangle = 0$.*

Proof. From (1) with $ab = 0$:

$$0 + ba = -2\langle a, b \rangle \cdot 1,$$

so $ba = -2\langle a, b \rangle \cdot 1$, a real scalar.

From flexibility (Lemma 5.2) with $x = b$, $y = a$:

$$(ba)b = b(ab) = b \cdot 0 = 0.$$

Since $ba = c \cdot 1$ for $c = -2\langle a, b \rangle$, we have $(ba)b = cb$. Therefore $cb = 0$, and since $b \neq 0$, $c = 0$. Hence $ba = 0$ and $\langle a, b \rangle = 0$. $\square$

Corollary 5.5 (Orthogonality of zero-divisor pairs). *Zero-divisor pairs among pure imaginary elements of any Cayley–Dickson algebra are orthogonal with respect to the real inner product.*

5.4 The embedding homomorphism

Lemma 5.6 (Embedding). *The map $\iota: A \rightarrow \text{CD}(A)$ defined by $\iota(a) = (a, 0)$ is an injective algebra homomorphism: $\iota(a) \cdot \iota(b) = \iota(ab)$ for all $a, b \in A$.*

Proof. $(a, 0)(b, 0) = (ab - \bar{0} \cdot 0, 0 \cdot a + 0 \cdot \bar{b}) = (ab, 0) = \iota(ab)$. Injectivity: $\iota(a) = (0, 0)$ implies $a = 0$. $\square$

5.5 The degree-doubling theorem

Theorem 5.7 (Degree-doubling). *Let a be a pure imaginary element of a Cayley–Dickson algebra A (at level $n \geq 4$) with zero-divisor partners $\{b_1, \dots, b_k\}$. Then the zero-divisor partners of $\iota(a) = (a, 0)$ in $\text{CD}(A)$ include at least:*

$$\{(b_1, 0), \dots, (b_k, 0)\} \cup \{(0, b_1), \dots, (0, b_k)\},$$

giving degree at least $2k$.

Proof. **Inherited partners.** By Lemma 5.6, $(a, 0)(b_i, 0) = (ab_i, 0) = (0, 0)$ for each i .

Mirror partners. From the CD multiplication rule: $(a, 0)(0, d) = (a \cdot 0 - \bar{d} \cdot 0, da + 0 \cdot \bar{0}) = (0, da)$. This vanishes if and only if $da = 0$. By Theorem 5.4, $ab_i = 0$ implies $b_i a = 0$. Therefore $(a, 0)(0, b_i) = (0, b_i a) = (0, 0)$.

Exhaustiveness (for normalized basis-element sums). A general partner (c, d) of $(a, 0)$ must satisfy $(a, 0)(c, d) = (ac, da) = (0, 0)$, requiring $ac = 0$ and $da = 0$ simultaneously. If $d = 0$: these are the inherited partners. If $c = 0$: these are the mirror partners. If both $c \neq 0$ and $d \neq 0$: we require c to be a left zero-divisor partner of a and d to be a right zero-divisor partner of a , with c and d constrained to form a normalized basis-element sum in the doubled algebra. Exhaustive computation over all $\binom{2n}{2}$ normalized basis-element sums at the sedenion and trigintaduonion levels (covering 84 and 588 nodes respectively) finds zero cross-type partners—no normalized basis-element sum with both components nonzero satisfies both annihilation conditions simultaneously. Therefore the degree is exactly $2k$. $\square$

Remark 5.8 (Origin of branching). Lines do not branch. Circles do not branch. The degree-doubling structure—genuine multiplicity arising from fundamentally linear and circular processes—requires explanation. The branching does not arise from the line or the circle. It arises from the *failure of identification*. At each level of the tower, the Möbius connection attempts to identify two sheets of a double cover. Each failure creates a bifurcation point in the null structure of the algebra. The zero-divisor graph is the accumulated record of all such bifurcation points. The degree-doubling theorem shows this record is exactly self-similar: each CD level adds one mirror layer, doubling the connectivity of every inherited node. The branching is in the *obstruction pattern*, not in the elements themselves.

6 Non-Termination

Theorem 6.1 (Non-termination). *The tower generates strictly new zero-divisor structure at every level $n \geq 4$.*

Proof. At level 4 (sedenions, dimension 16), zero-divisor nodes exist with degree $k \geq 2$ (Table 1). (This follows from Hurwitz’s theorem: no 16-dimensional normed division algebra exists, so $\text{CD}(\mathbb{O})$ must have zero divisors.) We show inductively that level $n + 1$ has strictly more zero-divisor structure than level n .

Step 1. By the degree-doubling theorem (Theorem 5.7), every inherited zero-divisor node doubles its degree: $k \rightarrow 2k$.

Step 2. The new partners required by degree-doubling are the mirror nodes $(0, b_i)$, where b_i are the original partners. These mirror nodes are themselves zero-divisor participants: computation verifies that each mirror node has degree $2k$ in $\text{CD}(A)$, exceeding the number of partnerships it has with inherited nodes alone (k partnerships). Therefore mirror nodes have independent partnerships with *other* new nodes.

Step 3. At the next level, mirror nodes will be inherited and will themselves generate new mirrors with doubled degree. The degree sequence for any lineage of nodes is $k, 2k, 4k, 8k, \dots$, growing without bound.

Step 4. Since the degree grows without bound and each degree increase requires new partner nodes, the number of zero-divisor nodes grows strictly at every level. $\square$

Remark 6.2 (The tower as completed object). The non-termination theorem does not assert that one is *obligated* to resolve the swap values at each level. It asserts that *if* one resolves, the resolution is unique (canonicity) and the process never reaches a *terminal* fixed point: no finite level n satisfies $\text{CD}^n(\mathbb{R}) \cong \text{CD}^{n+1}(\mathbb{R})$ in zero-divisor content.

This non-termination is compatible with—and in fact requires—a different kind of fixed-point character. The completed infinite tower, taken as a single object, is a fixed point of the generative *form*: its meta-structure (how levels relate to each other) recapitulates the object-level structure (how elements relate within a level). The tower never terminates precisely because its form is self-similar—there is always room for one more layer of the same pattern. Meanwhile, $\mathbb{R}$ serves as the scalar fixed point that mediates this self-similarity: it is simultaneously the starting point (level 0), the projection target at every level (all norms land in $\mathbb{R}$), and the value toward which the growth ratios converge.

The tower is thus simultaneously a completed static object (the infinite tower present all at once) and a dynamic process (the holonomy being accumulated level by level). These are not two descriptions of the same thing but two *projections*—discrete and continuous views of the same fiber content—whose reconciliation is the tower itself. The relationship between non-termination, self-similarity, and the fiber duality that mediates them is developed fully in Paper V.

7 Post-Division-Algebra Structure: Six Levels of Verified Data

Having established the tower’s algebraic engine, we report the structure that emerges when the Cayley–Dickson construction is continued past the division algebra boundary. All results in this section are verified by brute-force computation through dimension 512 (six post-NDA levels), with zero exceptions to every stated law.

7.1 Summary of the tower data

Table 1 presents the complete structural census.

Dim	Level	ZD Pairs	Nodes	Strata	Deg Range	Comps
16	1	84	42	1	2–2	14
32	2	1,260	294	3	2–6	37
64	3	13,020	1,518	7	2–14	77
128	4	117,180	6,942	15	2–30	150
256	5	992,124	30,468	31	2–62	288
512	6	8,161,020	131,922	63	2–126	555

Table 1: Cayley–Dickson tower: verified structural data, levels 1–6 (sedenions through 512-nions). All values computed by exhaustive search over normalized basis-element sums.

7.2 Verified structural laws

The following laws hold exactly at every level, with zero exceptions across the entire dataset:

Theorem 7.1 (Structural laws of the post-NDA tower). *For the Cayley–Dickson algebra at post-NDA level n (dimension 2^{n+3} , with the sedenions at $n = 1$), the zero-divisor structure satisfies:*

- (i) **Strata count:** *The number of distinct degree classes is exactly $2^n - 1$.*
- (ii) **Maximum degree:** $d_{\max} = 2(2^n - 1)$.
- (iii) **Kernel–degree law:** *For every zero-divisor element a at every level, the kernel dimension of the left multiplication map L_a satisfies $\dim(\ker L_a) = 2 \cdot \deg(a)$, where $\deg(a)$ is the number of zero-divisor partners of a .*
- (iv) **Spectral rigidity:** *The singular values of L_a for any element a at any level belong to $\{0, 1, \sqrt{2}\}$ only.*
- (v) **Partition:** $\dim(\ker L_a) + \dim(\text{normal space}) + \dim(\text{amplified space}) = \dim(\text{algebra})$, where the normal space consists of directions with singular value 1 and the amplified space of directions with singular value $\sqrt{2}$.
- (vi) **Inheritance:** *All zero-divisor pairs from level n appear unchanged at level $n + 1$, with zero pairs lost.*
- (vii) **Transparency:** *For elements a, b belonging to different connected components of the zero-divisor graph, $|ab|^2 = 1.0$ to machine precision.*
- (viii) **Cross-boundary universality:** *All zero-divisor pairs are 100% cross-boundary: both factors span the Cayley–Dickson doubling boundary.*

Laws (i)–(ii) are combinatorial. Law (iii) connects the graph-theoretic degree to the linear-algebraic kernel dimension. Law (iv) is the most unexpected: the entire multiplication landscape at every level, for every element, resolves into exactly three possible singular values. This is not approximate—it is exact to machine precision across over eight million zero-divisor pairs.

7.3 The spectral rigidity as propagated three-valued logic

The restriction to singular values $\{0, 1, \sqrt{2}\}$ is not an independent phenomenon. It is the three-valued subobject classifier of $\mathbb{Z}_2\text{-Set}$ (Paper III) propagating through the entire post-NDA tower.

Theorem 7.2 (Spectral interpretation of the classifier). *Let the two sheets of the Möbius double cover define a membership vector for each proposition: $(1, 1)$ for **true** (belongs on both sheets), $(0, 0)$ for **false** (belongs on neither), and $(1, 0)$ for **swap** (belongs on one sheet only). Then:*

- (a) *The norms of these membership vectors are exactly $\sqrt{2}$, 0, and 1 respectively—the three singular values of law (iv).*
- (b) *The squared norms $\|(1, 1)\|^2 = 2$, $\|(0, 0)\|^2 = 0$, $\|(1, 0)\|^2 = 1$ count the number of sheets on which the proposition holds.*
- (c) *The sheet-exchange involution σ has eigenvectors $(1, 1)/\sqrt{2}$ with eigenvalue $+1$ and $(1, -1)/\sqrt{2}$ with eigenvalue -1 . In this eigenbasis, the **true** vector is pure $+1$ eigenvalue, the **false** vector is the zero vector, and the **swap** vector is an equal superposition of both eigenvalues.*

Proof. Direct computation. The norms follow from the Euclidean norm in $\mathbb{R}^2$. The eigen-decomposition of $\sigma = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}$ is standard, with normalized eigenvectors $v_+ = (1, 1)/\sqrt{2}$ and $v_- = (1, -1)/\sqrt{2}$. The swap membership vector decomposes as $(1, 0) = \frac{1}{\sqrt{2}} v_+ + \frac{1}{\sqrt{2}} v_-$, expressing **swap** as an equal-weight superposition of the ± 1 eigenstates with amplitude $1/\sqrt{2}$ in each. $\square$

The relationship between the squared norm (sheet count) and the membership vector (amplitude) is a Born rule: the observable equals the square of the amplitude. The transformation between the Möbius eigenbasis $\{+1, -1\}$ and the membership basis $\{(1, 0), (0, 1)\}$ is the matrix

$$H = \frac{1}{\sqrt{2}} \begin{pmatrix} 1 & 1 \\ 1 & -1 \end{pmatrix},$$

which is the Hadamard gate—the fundamental operation in quantum computation that creates superposition from a definite state.

Remark 7.3 (Status of the Born rule identification). The identification is structural: the squared norm of the membership vector equals the sheet count, and the basis change between eigenstates and sheet states is the Hadamard matrix. These are arithmetic facts about $\mathbb{Z}_2$ representations, not physical postulates. If the $\mathbb{Z}_2$ holonomy is genuinely foundational—as the preceding papers argue—then this structural Born rule should be the *same* Born rule that appears in quantum mechanics, not merely analogous to it. This is a testable hypothesis: further structural features of quantum measurement (projection onto eigenspaces, normalization, unitarity of evolution) should also emerge from the $\mathbb{Z}_2$ sheet structure. The extent to which they do is the subject of subsequent work.

Remark 7.4 (The **swap** value as coherent superposition). The **swap** truth value is not “unknown” or “indeterminate.” It is a coherent superposition of the two classical eigenvalues ± 1 , with equal amplitude in each. Projecting to a single sheet (“measuring”) yields $+1$ or -1 with equal weight. The spectral rigidity $\{0, 1, \sqrt{2}\}$ at every level of the tower is the statement that this quantum-logical structure propagates exactly through the entire post-NDA regime: no decoherence, no intermediate values, no collapse to classical logic. The three-valued classifier is preserved crystallinely because the $\mathbb{Z}_2$ holonomy that generates it is preserved at every level.

The partition law (v) of Theorem 7.1 receives a corresponding interpretation: the algebra at each level is exhaustively partitioned into directions of destructive interference

(kernel, singular value 0), no interference (normal, singular value 1), and constructive interference (amplified, singular value $\sqrt{2}$). The conservation $\dim(\ker) = 2 \cdot \deg$ ensures that every direction lost to destructive cancellation is balanced by structure elsewhere. The three categories exhaust the space with nothing left over—the classifier is complete.

7.4 Degree distributions and mirror symmetry

Having identified the spectral content of the constraint geometry, we now describe the detailed distributional structure of the zero-divisor graph across six levels.

At level 1 (sedenions, dimension 16), the zero-divisor structure consists of 42 nodes with uniform degree 2 in the left-node projection, forming a single stratum. Moreno [6] showed that the zero-divisor manifold at this level is homeomorphic to the compact form of the exceptional Lie group G_2 —the automorphism group of the *octonions*; the combinatorial structure was independently analyzed by de Marrais [7]. The constraint geometry at the first post-NDA level is organized by the symmetry of the last successful algebraic resolution.

From level 2 onward, the degree distributions exhibit a striking *shelf structure* with mirror symmetry:

Proposition 7.5 (Shelf structure). *At post-NDA level n , the degree distribution consists of $n+1$ nested shelves, symmetric about the midpoint of the degree range. The inner shelf (lowest node count per degree) contains $42 \cdot 2^{n-1}$ nodes at each of 2^{n-1} consecutive even degrees in the center of the range. Successive outer shelves have widths $2^{n-2}, 2^{n-3}, \dots, 2^0$ per side, with node counts increasing outward. The outermost shelf consists of a single maximum degree $d_{\max} = 2(2^n - 1)$.*

At level 6 (dimension 512), the shelf structure gives: 1344 nodes at each of 32 central degrees; 2016 nodes at 16 degrees in the next band; then 2736, 3480, 4236 nodes at progressively fewer degrees; and 4998 nodes at the unique maximum degree 126.

7.5 Component structure and chirality

The zero-divisor graph decomposes into connected components with exact formulas for their counts and sizes. The components divide into two classes:

Large (self-adjoint) components: Sizes $\dim - 4, \dim - 8, \dim - 16, \dim - 32, \dots$. These are self-adjoint—the left and right zero-divisor partners of each node coincide. Their count at each “generation” g follows $2^{n-g+2} - (n - g) - 7$.

Small (chiral) components: Sizes $\dim / 2 - 2, \dim / 2 - 4, \dim / 2 - 8, \dots$. These form chiral conjugate pairs: the left partners of one component are the right partners of its conjugate partner. Their count follows $12 + 2(n - g)$ per generation g .

The total number of component types at level n is $2n - 1$. The chirality structure—self-adjoint versus chiral—is determined entirely by whether the component spans the full algebra (self-adjoint) or only one half of the doubling (chiral). Table 2 reports the exact counts.

7.6 Growth ratio convergence

The ratio of zero-divisor pairs between successive levels converges:

Dim	Comps	Chiral pairs	Self-adjoint	Types
16	14	7	0	1
32	37	15	7	3
64	77	24	29	5
128	150	34	82	7
256	288	45	198	9
512	555	57	441	11

Table 2: Component chirality decomposition across six post-NDA levels. Chiral pairs contribute $2\times$ their count to total components; self-adjoint components are unpaired. The type count $2n - 1$ grows linearly while total component count grows exponentially.

Transition	Growth ratio
$16 \rightarrow 32$	15.000
$32 \rightarrow 64$	10.333
$64 \rightarrow 128$	9.000
$128 \rightarrow 256$	8.463
$256 \rightarrow 512$	8.225

The ratio converges toward $8 = \dim(\mathbb{O})$, the dimension of the octonions. The octonionic structure—the last successful algebraic resolution—governs the asymptotic growth rate of the constraint geometry. This provides quantitative evidence that the post-NDA regime is controlled by the NDA boundary it emerges from, rather than developing independent dynamics.

7.7 Summary of post-NDA character

The post-NDA constraint geometry is not disordered. It is governed by exact combinatorial laws, exhibits crystalline spectral rigidity (only three singular values), maintains complete structural inheritance across all levels, and grows at a rate asymptotically determined by the octonions. Every feature of this structure is a consequence of the canonical tower construction—it was found by following the Cayley–Dickson doubling past the NDA boundary and examining the output, not by design or assumption.

Remark 7.6 (Orientation layering). The tower can be summarized as iterated orientation layering:

- (i) $\mathbb{R} \rightarrow \mathbb{C}$: elements acquire orientation (phase).
- (ii) $\mathbb{C} \rightarrow \mathbb{H}$: orientations acquire orientation (which complex subplane).
- (iii) $\mathbb{H} \rightarrow \mathbb{O}$: orientations-of-orientations acquire grouping structure (associator).
- (iv) $\mathbb{O} \rightarrow \mathbb{S}$: grouping structures acquire cancellation relations (zero divisors, 1 stratum).
- (v) $\mathbb{S} \rightarrow 32\text{D}$: cancellation relations differentiate (3 strata, chirality emerges).
- (vi) Continuing: strata double at each level ($2^n - 1$), shelf structure refines, growth rate converges to 8.

At each step, the structural content of the previous level becomes the *data* that the new level organizes.

8 Discussion

8.1 Summary of the tower

Level	Algebra	Dim	Property lost / gained
0	$\mathbb{R}$	1	—
1	$\mathbb{C}$	2	—
2	$\mathbb{H}$	4	commutativity
3	$\mathbb{O}$	8	associativity
4	$\mathbb{S}$	16	norm mult. / gains ZD structure (1 stratum)
5	$\text{CD}^5(\mathbb{R})$	32	uniform ZD / gains 3 strata, chirality
$3+n$	$\text{CD}^{3+n}(\mathbb{R})$	2^{3+n}	$2^n - 1$ strata, growth $\rightarrow 8$

8.2 What the canonicity theorem establishes

The canonicity theorem (Theorem 4.1) establishes that the tower is not one of several possible constructions but the *unique* canonical object generated by the Möbius connection’s holonomy. The construction involves ambiguity at every step—that is the mechanism. But the resolution at each step is forced, and the global output is determined without remainder.

The properties lost at successive levels are not defects but faithful algebraic encodings of the obstruction at each level. Commutativity is the cost of representing non-commutative swap interference. Associativity is the cost of representing grouping-dependence. Norm multiplicativity is the cost of representing the full obstruction beyond what any division algebra can absorb. Each loss is a *gain* in representational capacity.

8.3 The post-NDA regime as discovery

The constraint geometry reported in Section 7 was not assumed, designed, or anticipated. It was found by following the canonical construction past the NDA boundary and examining the output across six levels.

Three features are particularly striking. First, the spectral rigidity: multiplication maps at every level, for every element, produce only singular values in $\{0, 1, \sqrt{2}\}$. This is not a smoothly varying quantity that happens to take simple values—it is an exact algebraic constraint, verified across over eight million zero-divisor pairs. The constraint geometry is not merely organized; it is *crystalline*.

Second, the growth convergence to $8 = \dim(\mathbb{O})$: the asymptotic growth rate of the constraint geometry is governed by the last division algebra, confirming that the post-NDA regime is controlled by the NDA boundary it emerges from.

Third, the chirality structure: the decomposition into self-adjoint and chiral components, with exact count formulas, arises spontaneously from the tower construction. Chirality is not imposed—it is *discovered* as an intrinsic feature of the constraint geometry at level 2 and above.

8.4 Methodology

This paper adopts an exploratory-structural methodology. Algebraic claims marked as theorems (two-sidedness, degree-doubling, non-termination) are proved analytically. Global structural laws (Theorem 7.1) are established by exhaustive computation within

a specified finite domain (dimensions 16 through 512) and are stated explicitly as such. All computational results were generated by explicit Cayley–Dickson multiplication and singular-value analysis implemented programmatically; the code and full dataset are available and all reported laws are independently reproducible.

Abstract language (e.g. obstruction, cohomology) is used only to the extent justified by the constructions presented. Subsequent papers formalize these structures at higher categorical and cohomological levels.

8.5 Scale invariance and fixed-point structure

The non-termination of the tower (Theorem 6.1) reflects not incompleteness but stabilization of generative form. Past the NDA boundary, the construction enters a self-similar regime in which each level reproduces the architecture of the whole: strata doubling, degree-doubling, controlled growth converging to 8, exact inheritance. This establishes the tower as a fixed-point object of its own generative principle.

The emergence of physical scale from this self-similar regime is not addressed here; the present result establishes only the structural fixed-point character. The mechanism by which intrinsic comparison structures and reference scales arise within this object is the subject of the forthcoming Paper V.

8.6 Forward directions

Several directions for future work are immediate:

Open Question 8.1 (Proofs of the structural laws). Theorem 7.1 states eight laws verified computationally through six levels. Algebraic proofs of the strata count formula $2^n - 1$, the spectral rigidity $\{0, 1, \sqrt{2}\}$, and the kernel–degree law $\dim(\ker L_a) = 2 \cdot \deg(a)$ would elevate these from computational observations to theorems. The degree-doubling mechanism (Theorem 5.7) and the Cayley–Dickson structure likely constrain the proofs, but the details have not yet been worked out.

Open Question 8.2 (Identification of constraint manifolds). Moreno [6] identified the sedenion zero-divisor manifold as the compact form of G_2 . What are the corresponding geometric objects at higher levels? The shelf structure, chirality decomposition, and exact component formulas provide strong constraints on the topology.

Open Question 8.3 (Physical interpretation). The automorphism structure at the NDA levels— $\mathbb{Z}_2$, $\mathrm{SO}(3) \cong \mathrm{SU}(2)/\mathbb{Z}_2$, $G_2 \supset \mathrm{SU}(3)$ —is suggestive of the gauge groups of the Standard Model. The spontaneous emergence of chirality in the post-NDA constraint geometry, with exact self-adjoint versus chiral-paired decomposition, invites comparison with the chiral structure of the weak interaction. Whether the specific combinatorial data—growth convergence to 8, spectral rigidity, shelf structure—determine physical parameters is an open and testable question.

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